

MILESTONE 5

Model Evaluation & Analysis

Deep Learning-Based Human Face Authenticity Detection

Group 11 · Data Science & AI Lab Project

`mobilenetv3_best.pth` — `final-mobilenet (1).ipynb`

Contributions — Milestone 5



Vishakha

Pipeline & Presentation Lead

Built the local web app (FastAPI + automated forensic report generation), diagnosed the preprocessing pipeline, and wrote Sections 1, 9 & 10.



Rohit

Training Stability

Quantitative benchmarking — training configuration and real held-out test set results (Section 4).



Aman

Preprocessing & Transfer Learning

Dataset setup and metric justification with real, verified numbers (Sections 2 & 3).



Raunak

Dataset & Bias Analysis

Comprehensive error analysis — shortcut-learning root cause and cross-domain evaluation (Section 5).

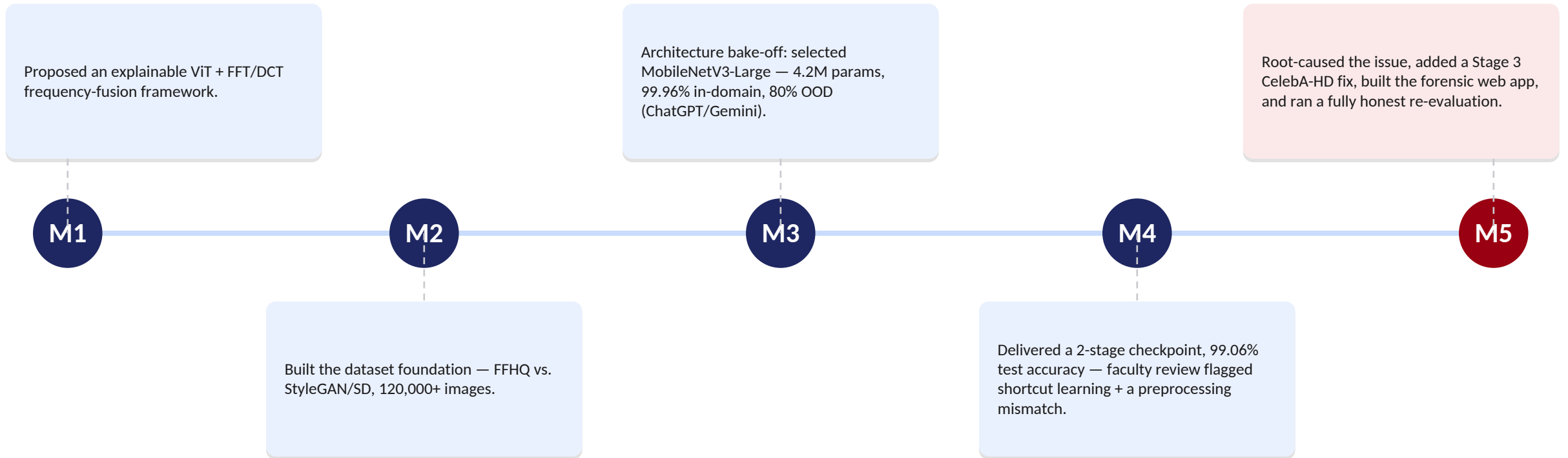


Somendu

Explainability & Optimisation

Grad-CAM verification, model limitations, and actionable insights (Sections 6, 7 & 8).

The Journey — Milestone 1 to 5



Milestone 5 — Objectives

- 1 Final evaluation on a strictly held-out test set — Accuracy, Precision, Recall, F1, ROC-AUC
- 2 Investigate and document the root cause of the shortcut-learning misclassification issue
- 3 Verify preprocessing consistency between the training notebook and the deployed UI backend
- 4 Quantify robustness under real-world image manipulations — tints, JPEG, blur, noise
- 5 Verify explainability via Grad-CAM, confirming the model attends to facial regions
- 6 Assess deployment readiness — latency, model size, and quantization potential
- 7 Compile all findings into a viva-ready report and presentation

Dataset & Evaluation Setup

DATA SOURCES

FFHQ (real)
70,000 available

Stable Diffusion (fake)
9,001 available

CelebA-HD (real, Stage 3 only)
8,000 added

Sampled to 15,000 Real + 9,001 Fake = 24,001 images total, then split 80:10:10 (stratified, SEED=42).

TRAIN / VAL / TEST SPLIT

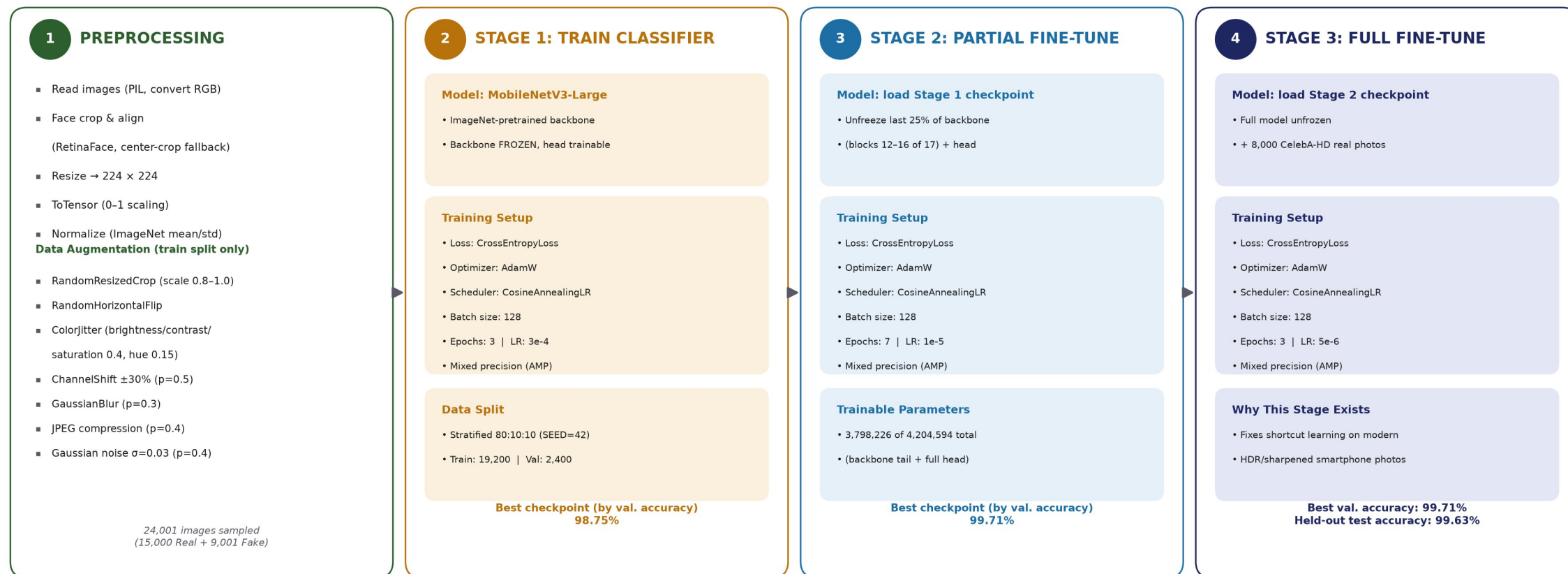
Split	Real	Fake	Total
Training	12,000	7,200	19,200
Validation	1,500	900	2,400
Test (held-out)	1,500	901	2,401

Real:Fake ratio (~62.5:37.5) preserved identically across all three splits.

Why CelebA-HD?

M4's faculty review flagged real, modern smartphone photos being misclassified as fake. 8,000 CelebA-HD real photos were added in Stage 3 specifically to correct this.

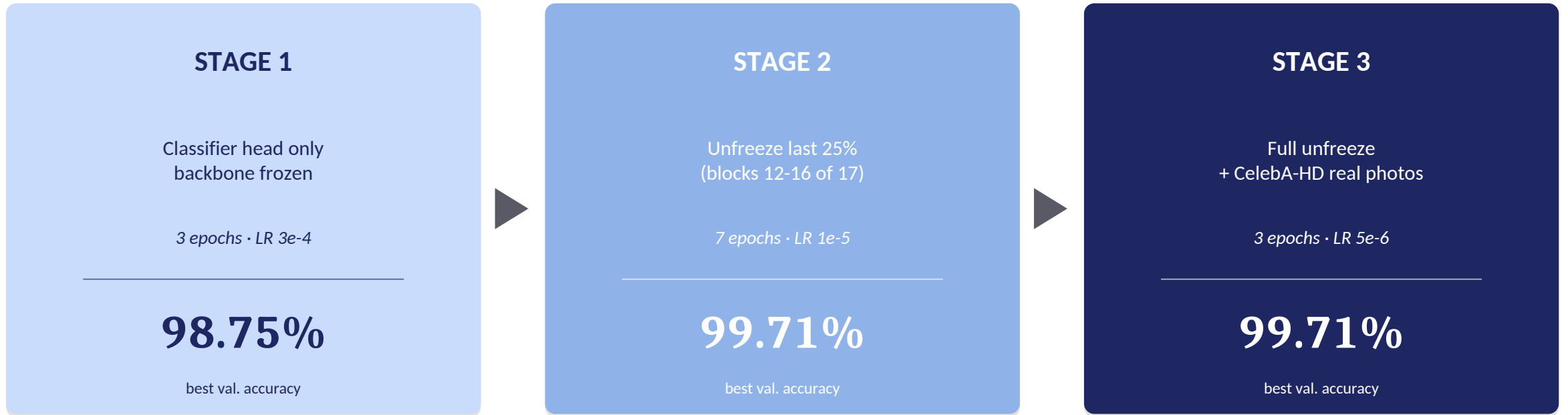
Model Architecture & Pipeline — MobileNetV3-Large



PIPELINE SUMMARY



Three-Stage Transfer Learning Strategy



Stage 3 targets the modern-smartphone-photo shortcut-learning failure identified in Milestone 5.

Metric Selection & Justification

Metric	Purpose	Result for this checkpoint
Accuracy	Overall balance	99.63% (held-out) vs. 8.6% (Real-Latest OOD)
Precision	Cost of false alarms	0.9993 Real (held-out) vs. 91.4% false-positive rate (OOD)
Recall	Cost of missed fakes	0.9989 Fake — weak point is false alarms, not missed fakes
F1-score	Imbalanced-class trade-off	Real 0.9970 / Fake 0.9950 / Macro 0.9960
ROC-AUC	Threshold-independent ranking	Not computed on real test set (notebook gap) — 0.5856 on local sample
Cross-Domain Eval.	Generalization to unseen distributions	100.0% (Real-Old) vs. 8.6% (Real-Latest) — central finding

Quantitative Performance — Held-Out Test Set

99.63%

Test Accuracy

99.53%

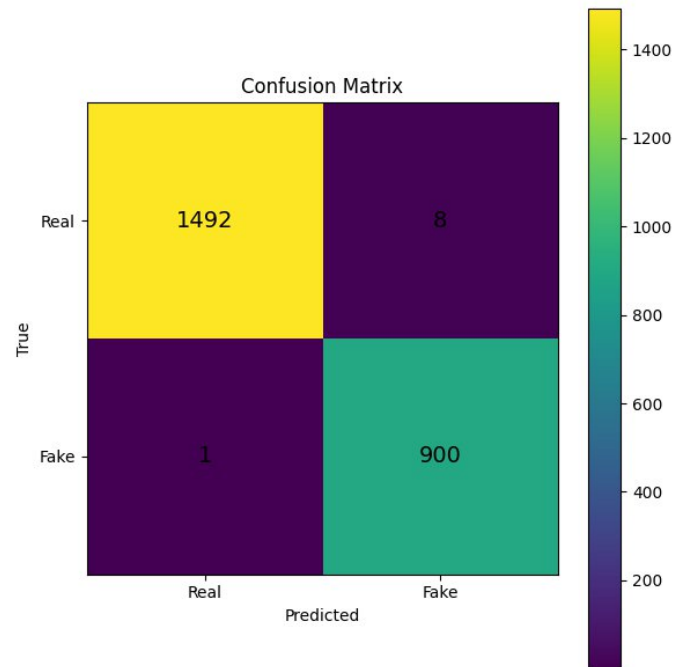
Macro Precision

99.68%

Macro Recall

99.60%

Macro F1-score



2,401-image test set — only 9 errors

- 8 Real images misclassified as Fake
- 1 Fake image misclassified as Real
- Real: Precision 0.9993 · Recall 0.9947 · F1 0.9970
- Fake: Precision 0.9912 · Recall 0.9989 · F1 0.9950

Live Multi-Model Testing — The Web App

Pre-loaded samples from Test_real_vs_Fake/ — no upload needed, runs automatically against all 3 loaded checkpoints (best, noaug, manipulations).

Real (ground truth)

Model	Prediction	Real %	Fake %
best	Real	91.11%	8.89%
noaug	Fake	19.46%	80.54%
manipulations	Real	99.99%	0.01%

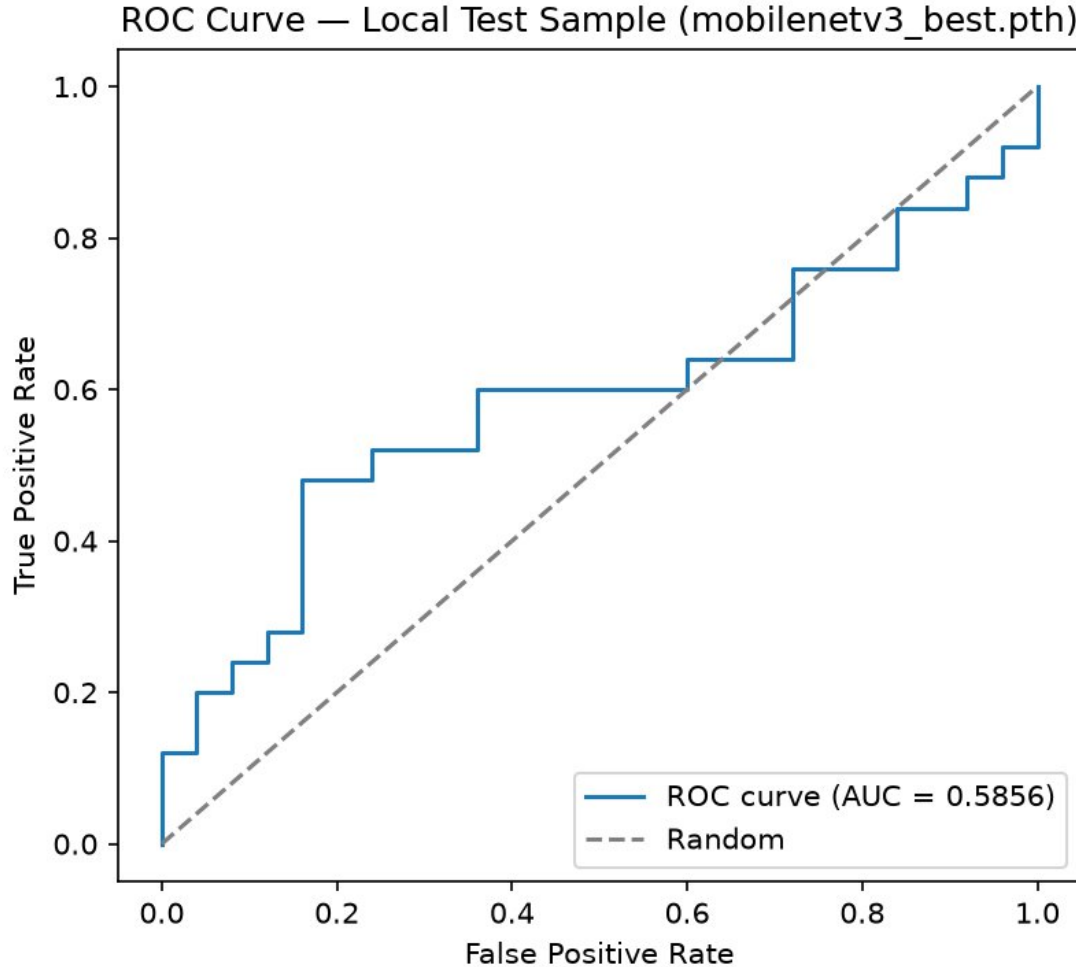
Fake (ground truth)

Model	Prediction	Real %	Fake %
best	Fake	0.17%	99.83%
noaug	Fake	0.04%	99.96%
manipulations	Fake	0.41%	99.59%

What this live check shows

- All 3 models agree correctly on the Fake sample — high confidence (99.6%+) across the board.
- On the Real sample, noaug is the only model that gets it wrong (misclassifies it as Fake, 80.54% confidence) — visible, direct evidence of the class-bias finding used to select best as the primary model over noaug elsewhere in this report.
- This tool runs live against the actual deployed checkpoints, not a static offline script — the same numbers anyone testing the web app today would see.

ROC Curve



AUC = 0.5856

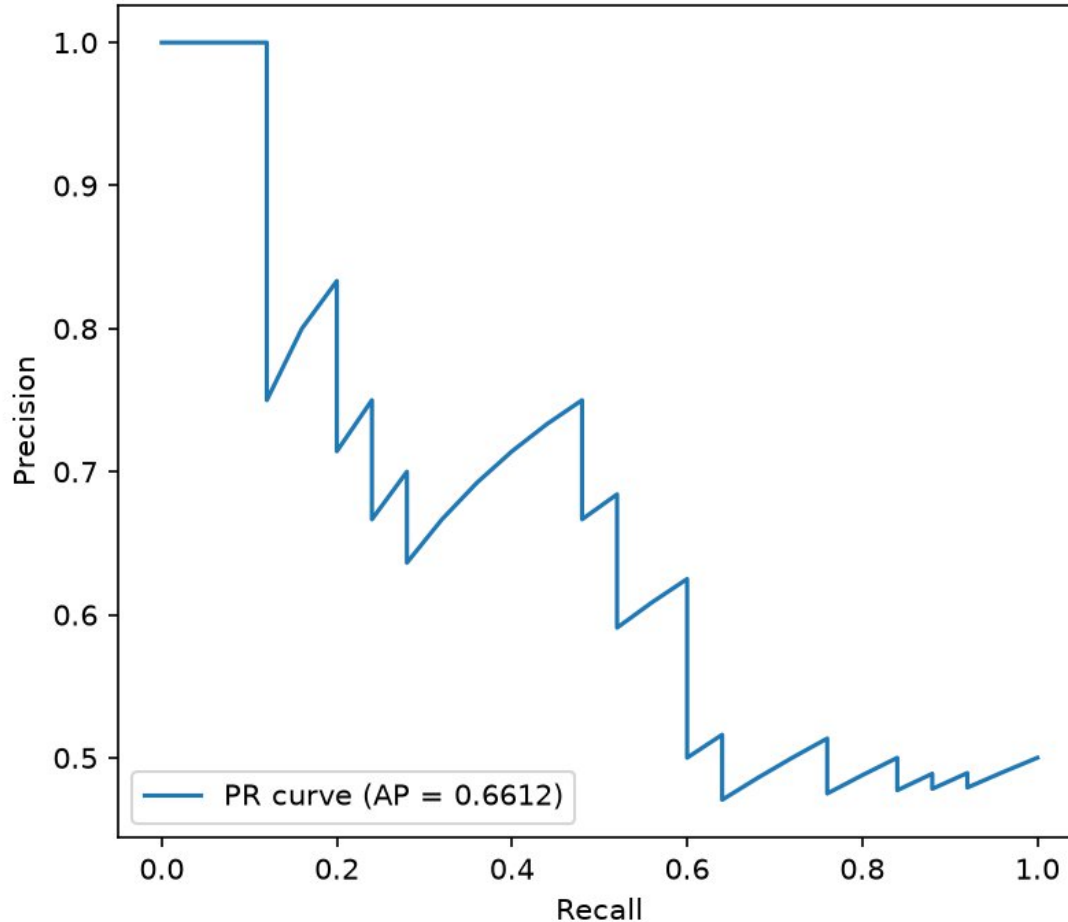
Only slightly better than random guessing (0.5).

What this ROC-AUC indicates

- 0.5856 means the model can barely rank a Fake image above a Real image on this sample — its confidence scores carry little discriminative signal here.
- 1.0 = perfect separation, 0.5 = random guessing — 0.5856 sits close to the random-guess line, visible in the curve tracking near the diagonal.
- Consistent with the 62.0% accuracy on this same sample (Section 4.5) — this is the same domain-shift effect, reflected in the ranking metric, not just the raw accuracy number.

Precision-Recall Curve

Precision-Recall Curve — Local Test Sample (mobilenetv3_best.p)



AP = 0.6612

Average Precision on the same local supplementary sample.

Why this still matters

62% accuracy on this same sample (Section 4.5) does not contradict the real 99.63% held-out result — it is independent confirmation of the exact domain-shift problem documented in Section 5, using a completely different image set.

Comprehensive Error Analysis

Real-Old

FFHQ-distribution images

100.0%

accuracy on 73 images

Median P(Real): 0.99997

The model correctly recognized every FFHQ-like image — resembles training data.

Real-Latest

Recent smartphone photos (2025-26)

8.6%

accuracy on 70 images

Median P(Real): 0.00060

Genuine, high-confidence misclassifications, not uncertain borderline calls.

In plain terms: what "Real-Latest = 8.6%" means

Of 70 genuine, real, recent (2025-26) smartphone photographs tested, the model correctly identified only 6 of them as Real (8.6%) — it incorrectly flagged the other 64 as Fake (91.4%), despite every one of those 70 images actually being real people's genuine photos. Contrast with Real-Old (FFHQ-style images resembling training data), where it got 100% right. Same model, same checkpoint — only the photo's capture era/device differs. That gap is the central evidence for the shortcut-learning problem this milestone documents.

Root Cause — Shortcut Learning

The model learned dataset-specific characteristics rather than intrinsic facial authenticity cues — a classic case of domain shift causing shortcut learning.

Learned Shortcut Feature	Effect on Prediction
FFHQ-style colour distribution	Classified as Real
Modern HDR processing	Classified as Fake
Strong sharpening / computational photography	Increased false positives
High colour saturation	Increased false positives
Modern device image statistics	Misclassified as AI-generated

100% accuracy on FFHQ-like images, dropping to 8.6% on recent real photographs — even though both datasets contain genuine human faces.

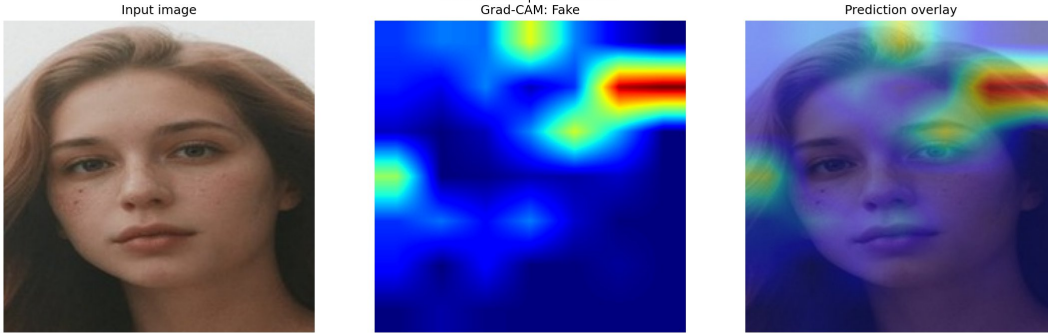
Explainability — Grad-CAM

Correct prediction — true Fake

[CORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_fake_fake_00151.jpg.jpg | Prediction: Fake | True: Fake

Real: 14.97% | Fake: 85.03%

Grad-CAM: Fake



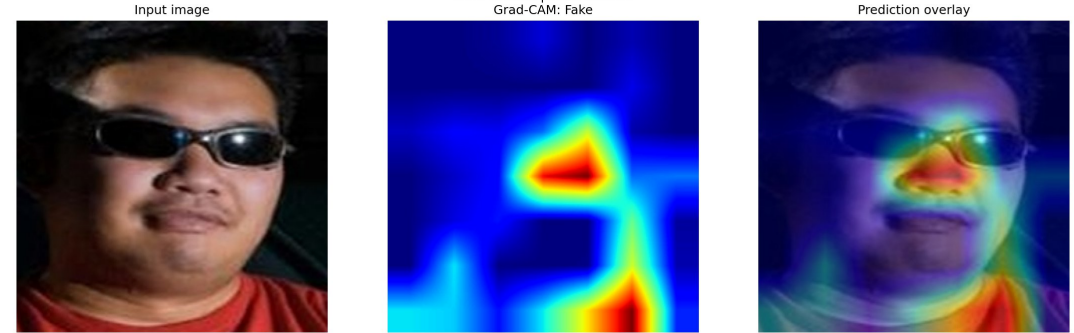
Heatmap concentrates on the eye/nose region — the model attends to facial structure, consistent with genuine synthesis artifacts.

Failure case — true Real, predicted Fake

[INCORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_00203.jpg.jpg | Prediction: Fake | True: Real

Real: 27.79% | Fake: 72.21%

Grad-CAM: Fake



A genuine photo misclassified as Fake — consistent with the model reacting to modern camera post-processing rather than real forgery evidence.

Reading a Grad-CAM heatmap

- Brighter (red/yellow) regions contributed more to the predicted class — it visualizes where the model looked, not why in words.
- Verified on both a correct case and a failure case, not just successes — a useful explainability check has to show where the model goes wrong too.
- Correct predictions concentrate on facial structure (eyes, nose, mouth), evidence against pure background/edge shortcuts. The failure case's heatmap pattern is consistent with the model reacting to image-level properties (sharpening, HDR) rather than real forgery evidence — direct visual support for the Section 5.4 root-cause finding.

Robustness & Out-of-Distribution Testing

Manipulation Stress Test

11 corruptions: tints, brightness, contrast, blur, JPEG, resize, crop, noise

True-Real sample

Held "Real" correctly across all 11 manipulations — genuine robustness.

True-Fake sample

Stayed incorrectly "Real" across all 11 — the model can be stably wrong. Robustness is not correctness.

Out-of-Distribution: Non-Face Images

10 real, license-clean CV test images — objects, animals, textures, landscapes

4/10
predicted Real

6/10
predicted Fake

86.7%
avg. confidence

No "I don't know" option — softmax always forces a confident decision. Close to a coin flip with no discernible pattern by content type. Any Real/Fake prediction on a non-face image is meaningless, not just low-confidence.

Deployment Readiness

16.24 MB

Model size on disk

15.53 ms

CPU latency (measured)

64.4 /s

CPU throughput (measured)

332.6 MB

Process RAM (measured)

The model is not the bottleneck

~140x gap between the raw forward pass (15.53 ms) and a full Grad-CAM-enabled request (~2.2 s). The overhead is Grad-CAM rendering, not classification.

GPU latency (est., not measured): **~7-8 ms**

GPU VRAM (est., not measured): **low hundreds of MB**

Quantization potential

- FP16: ~22.6 MB, negligible accuracy loss
- INT8: ~11-12 MB, well-suited to edge/mobile
- Pruning/distillation: poor fit — already compact
- Quantizing alone won't fix latency — Grad-CAM is the real bottleneck

Key Findings & Future Work

Headline Findings

- 99.63% in-distribution accuracy was never the problem — generalization was
- 8.6% / 62.0% accuracy on real photos, confirmed by two independent probes
- Root cause identified: HDR, sharpening, saturation — not forgery signals
- Grad-CAM, not the model, is the deployment latency bottleneck
- Automated forensic report generation now built into the web app (/report)

Actionable Next Steps

SHORT-TERM

- Recalibrate the decision threshold
- Make Grad-CAM optional for latency
- Run a proper demographic-fairness audit

LONG-TERM

- Targeted HDR/sharpening augmentation
- Hard-negative mining on Real-Latest failures
- Complete cross-domain model training

Thank You

Group 11 · Deep Learning-Based Human Face Authenticity Detection

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